MATH 202 Your name: _____

Learning target T3, version 1

Use the Ratio Test to determine the radius of convergence and (open) interval of convergence of a given power series.

Find the radius of convergence and the open interval of convergence for the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n \left(x+\frac{1}{3}\right)^n}{n \cdot 2^n}.$$

MATH 202 Your name: _____

Learning target T3, version 2

Use the Ratio Test to determine the radius of convergence and (open) interval of convergence of a given power series.

Find the radius of convergence and the open interval of convergence for the series

$$\sum_{n=1}^{\infty} \frac{3^n \left(x - \frac{1}{2}\right)^n}{n}.$$